

# Stirling Engine Efficiency Project

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## Abstract

The purpose of this lab is to experimentally determine the efficiency of a Stirling Engine and compare it with the efficiency predicted by the theoretical model. The Stirling Engine provided will be attached to a cart, meaning that it will need to produce both translational and rotational work to travel 3.05 m. The experimental results of the efficiency for the Stirling Engine are then compared, and as expected, it is much lower than the theoretical efficiency, due to only a small fraction of the combustion heat actually reaching the working gas, major heat loss, and various frictional losses in the system, as well as measurement uncertainties in fuel mass, timing, and cart motion.

## Background

A Stirling Engine operates based on a hot and cold side, where there is a vast temperature difference, and the trapped air is continuously cooled and heated, allowing the engine to generate work.

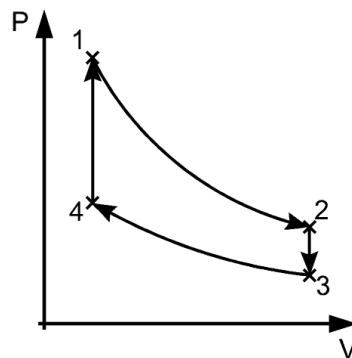


Figure 1. PV Diagram of the Stirling Cycle.

In terms of the thermodynamic cycle, the gas in an ideal Stirling Engine starts in stage 1, and is isothermally expanded while it moves to stage two. The gas remains at a constant temperature due to a pressure decrease, which cools the gas, and the heat from the heat source

keeps it constant. This is the process where the Stirling Engine generates work. In stages 2-3, the gas then experiences an isochoric cooling, where the hot gas travels to the cold side to be cooled down, releasing its heat without a volume change. In stages 3-4, the gas goes through an isothermal compression, where the temperature remains constant again due to the cold side cooling it down while the compression generates heat. Note that work is done on the gas in this stage. Lastly, the engine returns to stage 1 with an isochoric heating, where the gas returns to the hot side to be heated to restart the cycle without a volume change.

In this experiment, the  $T_H$  side is supplied from an alcohol lamp, while the  $T_C$  side is supplied from the temperature of the room. Based on this, the theoretical efficiency of the Stirling Engine can be calculated through the formula:

$$\eta_{Stirling} = 1 - \frac{T_C}{T_H}$$

The theoretical efficiency of the Stirling Engine is the same as the Carnot efficiency, as the Carnot efficiency is the maximum efficiency of an engine that operates between two temperatures. Experimentally, the efficiency must be calculated through work input and output. Work input is found through the energy received by the Stirling Engine from the combustion of the denatured alcohol in the alcohol lamp, while the work output is determined through measuring the data for the translational work of the cart and the rotational work of the four wheels that make up the model Stirling Engine. Therefore, based on the construction of the cart, the work output can be calculated using the equation:

$$W = \frac{1}{2}mv^2 + 4 * (\frac{1}{2}I\omega^2)$$

Where  $W$  is the total work done,  $m$  is the total mass of the cart,  $v$  is the velocity of the cart,  $I$  is the moment of inertia of a single wheel, and  $\omega$  is the angular velocity. The velocity will be determined from time and displacement, the moment of inertia of the wheels will be determined through CAD software due to the complex shape of the wheel, and the angular velocity will be determined from the wheels' radii, along with the velocity. This formula will allow us to estimate the work done over a period of time. Along with work in through the heat of the alcohol lamp, an experimental efficiency can be calculated using the equation:

$$\eta_{Experimental} = \frac{W_{out}}{W_{in}}$$

**Equipment:**

- Lab Essentials:
  - Stirling Engine on a cart
  - Lighter
- Other Complementary Equipment:
  - Marker
  - Ruler
  - Calipers
  - Temperature Probe
  - Stopwatch
  - Digital Scale

**Procedures:**

The experimental procedures began with obtaining a measurement of room temperature, along with the combustion temperature of alcohol to be used in the calculations for theoretical efficiency. Then the experimental track for the cart was set up to be labeled at 3.05 meters, and a stopwatch was prepared to record the time the Stirling Engine cart traveled the whole distance of the track. The alcohol lamp of the Stirling Engine was then filled with denatured alcohol, and the mass of the cart was recorded before and after 10 runs to obtain measurements of how much fuel was spent. After filling the alcohol lamp, the wick was lit, and the Stirling engine was held in place to reach full speed; the cart was placed on the track and timed until it reached the 3.05-meter mark, and then the flame was extinguished. For a test trial, a dot was marked on the wheel, and the wheel made approximately 17-18 revolutions over 305 cm, which indicates little to no slippage using circumference calculations. This experiment is then repeated 10 times.

## Results

Mass of wheel:  $20.23 \pm 0.001 \text{ g} = 0.02023 \pm 0.000001 \text{ kg}$

Moment of inertia of wheels: Calculated using CAD software. (Lzz is utilized due to the axis of rotation)

$$- 9848.771 \pm 100 \text{ g*mm}^2 = 9.8488*10^{-6} \pm 1*10^{-7} \text{ kg*m}^2$$

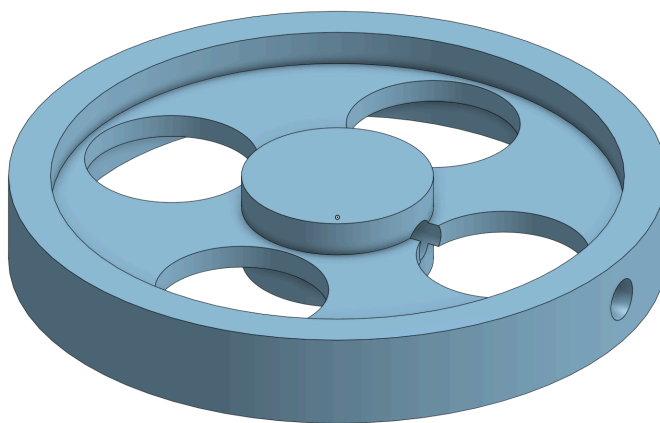


Figure 2. Image of CAD Software Designed Wheel.

| Mass moments of inertia (g mm <sup>2</sup> ) <input type="checkbox"/> Override |          |     |          |     |          |
|--------------------------------------------------------------------------------|----------|-----|----------|-----|----------|
| Lxx                                                                            | 4976.553 | Lxy | 0.002    | Lxz | 1.846e-4 |
| Lyx                                                                            | 0.002    | Lyy | 5051.598 | Lyz | 0.003    |
| Lzx                                                                            | 1.846e-4 | Lzy | 0.003    | Lzz | 9848.771 |

Figure 3. Image of Mass Properties of Wheel.

Diameter of wheel:  $54.8 \pm 0.05 \text{ mm} = 0.0548 \pm 5*10^{-5} \text{ m}$

Room temperature:  $21 \pm 0.5 \text{ }^{\circ}\text{C} = 294.15 \pm 0.5 \text{ K}$

Expected burning temperature of denatured alcohol:  $1920 \text{ }^{\circ}\text{C} = 2193.15 \text{ K}$

| Table 1: Rolled Distance, Time, and Final Mass of Stirling Engine                                                                                                                                                                                                                                                                                                                                                                                                    |                |                     |                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------|---------------------|
| Distance                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Time (s)       | Initial Mass (kg)   | Final Mass (kg)     |
| $3.05 \pm 0.05$ m                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $9.13 \pm 0.3$ | $0.4280 \pm 0.0001$ | $0.4273 \pm 0.0001$ |
| $3.05 \pm 0.05$ m                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $6.85 \pm 0.3$ | $0.4273 \pm 0.0001$ | $0.4271 \pm 0.0001$ |
| $3.05 \pm 0.05$ m                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $6.56 \pm 0.3$ | $0.4271 \pm 0.0001$ | $0.4267 \pm 0.0001$ |
| $3.05 \pm 0.05$ m                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $8.25 \pm 0.3$ | $0.4267 \pm 0.0001$ | $0.4264 \pm 0.0001$ |
| $3.05 \pm 0.05$ m                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $6.82 \pm 0.3$ | $0.4264 \pm 0.0001$ | $0.4262 \pm 0.0001$ |
| $3.05 \pm 0.05$ m                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $6.44 \pm 0.3$ | $0.4262 \pm 0.0001$ | $0.4168 \pm 0.0001$ |
| $3.05 \pm 0.05$ m                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $6.69 \pm 0.3$ | $0.4168 \pm 0.0001$ | $0.4165 \pm 0.0001$ |
| $3.05 \pm 0.05$ m                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $7.31 \pm 0.3$ | $0.4165 \pm 0.0001$ | $0.4162 \pm 0.0001$ |
| $3.05 \pm 0.05$ m                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $7.22 \pm 0.3$ | $0.4284 \pm 0.0001$ | $0.4281 \pm 0.0001$ |
| $3.05 \pm 0.05$ m                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $7.44 \pm 0.3$ | $0.4281 \pm 0.0001$ | $0.4277 \pm 0.0001$ |
| <p>Table 1: The distance traveled was a set distance marked using a tape measure and a marker; the time elapsed was obtained using a stopwatch, with its uncertainty being the human reaction speed; the initial mass of the cart was taken when the engine began running before being placed on the track, and the final mass was recorded after the cart had traveled the distance; their uncertainties use the measurement uncertainty for the digital scale.</p> |                |                     |                     |

## Analysis

| Table 2: Translational Work Done                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                     |                     |                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------|------------------------|
| Trials                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Mass (kg)           | Velocity (J)        | Translational Work (J) |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $0.4280 \pm 0.0001$ | $0.3341 \pm 0.0123$ | $0.0239 \pm 0.0009$    |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $0.4273 \pm 0.0001$ | $0.4453 \pm 0.0208$ | $0.0424 \pm 0.0020$    |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $0.4271 \pm 0.0001$ | $0.4649 \pm 0.0226$ | $0.0462 \pm 0.0022$    |
| 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $0.4267 \pm 0.0001$ | $0.3697 \pm 0.0147$ | $0.0292 \pm 0.0012$    |
| 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $0.4264 \pm 0.0001$ | $0.4472 \pm 0.0210$ | $0.0426 \pm 0.0020$    |
| 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $0.4262 \pm 0.0001$ | $0.4736 \pm 0.0234$ | $0.0478 \pm 0.0024$    |
| 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $0.4168 \pm 0.0001$ | $0.4559 \pm 0.0218$ | $0.0433 \pm 0.0021$    |
| 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $0.4165 \pm 0.0001$ | $0.4172 \pm 0.0184$ | $0.0363 \pm 0.0016$    |
| 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $0.4284 \pm 0.0001$ | $0.4224 \pm 0.0189$ | $0.0382 \pm 0.0017$    |
| 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $0.4281 \pm 0.0001$ | $0.4099 \pm 0.0178$ | $0.0360 \pm 0.0016$    |
| <p>Table 2: The mass values use the initial mass of the cart due to there being only a small difference in mass across the run time, with its uncertainty being the measurement uncertainty in the scale. The velocity values were calculated using time and displacement using the formula <math>v = \frac{\Delta x}{\Delta t}</math>, and their uncertainty is determined using the formula</p> $uv = \sqrt{\left(\frac{1}{t} * ux\right)^2 + \left(-\frac{x}{t^2}ut\right)^2}.$ <p>The translational work values were determined from the formula <math>W_T = \frac{1}{2}mv^2</math>, and their uncertainty is from the formula</p> $uW_T = \sqrt{\left(\frac{1}{2}v^2 * um\right)^2 + \left(\frac{1}{2}mv * uv\right)^2}.$ |                     |                     |                        |

| Table 3: Rotational Work Done                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                              |                             |                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-----------------------------|---------------------|
| Trials                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Moment of Inertia<br>(kg * m <sup>2</sup> )  | Angular Velocity<br>(rad/s) | Rotational Work (J) |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 9.8488*10 <sup>-6</sup> ± 1*10 <sup>-7</sup> | 0.0239 ± 0.00003            | 0.0029 ± 0.0013     |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 9.8488*10 <sup>-6</sup> ± 1*10 <sup>-7</sup> | 0.0424 ± 0.00007            | 0.0052 ± 0.0040     |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 9.8488*10 <sup>-6</sup> ± 1*10 <sup>-7</sup> | 0.0462 ± 0.00008            | 0.0057 ± 0.0047     |
| 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 9.8488*10 <sup>-6</sup> ± 1*10 <sup>-7</sup> | 0.0292 ± 0.00004            | 0.0036 ± 0.0019     |
| 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 9.8488*10 <sup>-6</sup> ± 1*10 <sup>-7</sup> | 0.0426 ± 0.00007            | 0.0052 ± 0.0040     |
| 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 9.8488*10 <sup>-6</sup> ± 1*10 <sup>-7</sup> | 0.0478 ± 0.00009            | 0.0059 ± 0.0050     |
| 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 9.8488*10 <sup>-6</sup> ± 1*10 <sup>-7</sup> | 0.0433 ± 0.00008            | 0.0055 ± 0.0043     |
| 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 9.8488*10 <sup>-6</sup> ± 1*10 <sup>-7</sup> | 0.0363 ± 0.00006            | 0.0046 ± 0.0031     |
| 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 9.8488*10 <sup>-6</sup> ± 1*10 <sup>-7</sup> | 0.0382 ± 0.00006            | 0.0047 ± 0.0032     |
| 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 9.8488*10 <sup>-6</sup> ± 1*10 <sup>-7</sup> | 0.0360 ± 0.00006            | 0.0044 ± 0.0029     |
| <p>Table 3: The moment of inertia values were determined using CAD software. The angular velocity values were determined using the formula <math>\omega = \frac{v}{r}</math>, and their uncertainty is found through the formula <math>u\omega = \sqrt{(\frac{1}{r} * uv)^2 + (-\frac{v}{r^2}ur)^2}</math>. The rotational work values were calculated using the formula <math>W_R = 4(\frac{1}{2}I\omega^2)</math>, and their uncertainty was calculated using the formula <math>uW_R = \sqrt{(2\omega^2 * uI)^2 + (2I\omega * u\omega)^2}</math>.</p> |                                              |                             |                     |

Based on these values, the experimental efficiency can then be calculated using the average of the total work, the sum of the translational and rotational work done, across 10 trials, along with the average work into the system using the mass difference and the heat of combustion of denatured alcohol, which is around 27000000 J/kg. This value is then compared to the theoretical efficiency of the engine, which is calculated through the temperatures, in Kelvin, of the hot and cold sides of the Stirling Engine model.

| Table 4: Velocity and Experimental Work Done |                   |                  |               |
|----------------------------------------------|-------------------|------------------|---------------|
| Avg. Tot. Work Done                          | Experimental Eff. | Theoretical Eff. | Percent Error |
| 0.0433                                       | 0.0052%           | 86.5978%         | 99.9940%      |

Table 4: The average total work done was found by summing the translational and rotational work of each of the trials and obtaining the average of them. The experimental efficiency was obtained using the formula  $\eta_{Experimental} = \frac{W_{out}}{W_{in}}$ , the theoretical efficiency was calculated using the formula  $\eta_{Stirling} = 1 - \frac{T_c}{T_H}$ , and the percent error was calculated using the formula

$$\% Error = \frac{|\eta_{Experimental} - \eta_{Stirling}|}{\eta_{Stirling}} * 100.$$

| velocity trial   | delta x        | delta t       | v              | v uncert       | mass diff (kg) | heat of comb (J/kg) | work in (J) |
|------------------|----------------|---------------|----------------|----------------|----------------|---------------------|-------------|
| 1                | 3.05           | 9.13          | 0.3341         | 0.0123         | 1              | 0.00003             | 810         |
| 2                | 3.05           | 6.85          | 0.4453         | 0.0208         | 2              | 0.00002             | 540         |
| 3                | 3.05           | 6.56          | 0.4649         | 0.0226         | 3              | 0.00004             | 1080        |
| 4                | 3.05           | 8.25          | 0.3697         | 0.0147         | 4              | 0.00003             | 810         |
| 5                | 3.05           | 6.82          | 0.4472         | 0.0210         | 5              | 0.00002             | 540         |
| 6                | 3.05           | 6.44          | 0.4736         | 0.0234         | 6              | 0.00004             | 1080        |
| 7                | 3.05           | 6.69          | 0.4559         | 0.0218         | 7              | 0.00003             | 810         |
| 8                | 3.05           | 7.31          | 0.4172         | 0.0184         | 8              | 0.00003             | 810         |
| 9                | 3.05           | 7.22          | 0.4224         | 0.0189         | 9              | 0.00003             | 810         |
| 10               | 3.05           | 7.44          | 0.4099         | 0.0178         | 10             | 0.00004             | 1080        |
| work trans trial | mass           | v             | work           | w uncert       |                |                     |             |
| 1                | 0.428          | 0.3341        | 0.0239         | 0.0009         |                |                     |             |
| 2                | 0.4273         | 0.4453        | 0.0424         | 0.0020         |                |                     |             |
| 3                | 0.4271         | 0.4649        | 0.0462         | 0.0022         |                |                     |             |
| 4                | 0.4267         | 0.3697        | 0.0292         | 0.0012         |                |                     |             |
| 5                | 0.4264         | 0.4472        | 0.0426         | 0.0020         |                |                     |             |
| 6                | 0.4262         | 0.4736        | 0.0478         | 0.0024         |                |                     |             |
| 7                | 0.4168         | 0.4559        | 0.0433         | 0.0021         |                |                     |             |
| 8                | 0.4165         | 0.4172        | 0.0363         | 0.0016         |                |                     |             |
| 9                | 0.4284         | 0.4224        | 0.0382         | 0.0017         |                |                     |             |
| 10               | 0.4281         | 0.4099        | 0.0360         | 0.0016         |                |                     |             |
| work rot trial   | i              | omega         | work           | w uncert       | omega uncert   | radius              |             |
| 1                | 0.000009848771 | 12.19209952   | 0.0029         | 0.0013         | 0.4483         | 0.0274              |             |
| 2                | 0.000009848771 | 16.2501998    | 0.0052         | 0.0040         | 0.7605         |                     |             |
| 3                | 0.000009848771 | 16.96857753   | 0.0057         | 0.0047         | 0.8249         |                     |             |
| 4                | 0.000009848771 | 13.49259013   | 0.0036         | 0.0019         | 0.5388         |                     |             |
| 5                | 0.000009848771 | 16.32168161   | 0.0052         | 0.0040         | 0.7668         |                     |             |
| 6                | 0.000009848771 | 17.28478221   | 0.0059         | 0.0050         | 0.8542         |                     |             |
| 7                | 0.000009848771 | 16.63884434   | 0.0055         | 0.0043         | 0.7950         |                     |             |
| 8                | 0.000009848771 | 15.22761541   | 0.0046         | 0.0031         | 0.6735         |                     |             |
| 9                | 0.000009848771 | 15.41743333   | 0.0047         | 0.0032         | 0.6892         |                     |             |
| 10               | 0.000009848771 | 14.96154148   | 0.0044         | 0.0029         | 0.6518         |                     |             |
| avg work done    |                | avg energy in | efficiency (%) | theo. eff. (%) | % error        |                     |             |
| 0.0433           |                | 837           | 0.0052         | 86.5878        | 99.9940        |                     |             |

Figure 2. Image of Excel Calculations

## Conclusion

In conclusion, this experiment showed the difference in the experimental and theoretical efficiency of a Stirling Engine. The lab process utilized a model Stirling Engine on a cart, where the source of heat was provided by an alcohol lamp burning denatured alcohol, with the cool side being the environmental temperature, which was converted into translational and rotational work for the cart to move. Over a measured distance of 3.05 m, the time the cart took to travel the distance was recorded with a stopwatch. The mass of the fuel spent was also measured for each trial by taking the initial mass and final mass of the cart after it completed the whole distance of the track. From this data, the velocity was determined, and using the velocity, the translational and rotational work done by the Stirling Engine was calculated. From these values, the average total mechanical work of 0.0433 J was obtained. Using the heat of combustion, 27000000 J/kg, and the burned mass of the denatured alcohol, we found an experimental efficiency of 0.0052%, while the theoretical efficiency of the Stirling engine was calculated to be 86.5978%. This gave a large percent error of 99.9940%, showing that there is a drastic difference between the efficiency of the model Stirling Engine and an ideal Stirling Engine.

To explain this large error between experimental and expected efficiency values, the values and constants used during the lab must be reexamined. First, for the experimental efficiency, the temperature of burning denatured alcohol, being 1920 °C, is an overestimate, as this temperature is only reached in perfect, oxygen-rich conditions. Instead, a more realistic temperature should be used in atmospheric conditions, where it burns at 1200-1500 °C, which is at the hottest part of the flame, and the fact that this temperature does not reach the engine is also not accounted for. Along with this, the time allotted for the engine to start up would only allow the air in the Stirling Engine to reach a fraction of the temperature of the flame.

For the experimental efficiency, the complete heat of combustion for the spent denatured alcohol is used, which is not reasonable. In reality, only a very small amount of the heat from the combustion actually reaches the working gas that becomes useful work input to the Stirling cycle, while all other heat energy is lost to the surroundings and within the system. For the Stirling Engine setup provided, an estimated maximum of 5% of the heat of combustion of the denatured alcohol being input as useful work into the Stirling Engine would be more realistic. During the lab process, the engine also took a long time to reach a steady speed, which caused fuel loss that is not accounted for, plus the continuous friction in the moving components of the engine and rolling resistance in the wheels. Further, the uncertainty in the time measurements also created errors in the lab values, and the track distance itself had measurement uncertainty; the cart did not consistently travel in a perfectly straight line from finish to end, so there was some additional distance that was traveled, since a diagonal line is longer than a straight line. Since the actual distance traveled was greater than the measured 3.05 m length, it caused us to underestimate the displacement used in our velocity and work calculations.

Ultimately, this lab showed the difference between experimental and theoretical efficiencies in a Stirling Engine through model and mathematical calculations. Although a large percentage error in the two efficiencies was calculated, careful examination of the overly optimistic assumptions allowed for the errors to be explained.

## Citations

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